

# RANDOM PROJECTIVE-HYPERPLANE TOGGLING IS A DISJOINT UNION OF CROWN WALKS

ANONYMOUS

ABSTRACT. Let  $E = \mathbb{F}_2^d \setminus \{0\}$ , and repeatedly toggle the points in the kernel of a uniformly sampled nonzero linear form. We determine this chain on the full  $2^{2^d-1}$ -state subset carrier. Its communicating classes are  $2^{2^d-d-2}$  equal cosets, and every class is simple random walk on the crown graph  $K_{2^d, 2^d}$  with one perfect matching removed. Fourier inversion gives the exact number of ordered histories between every pair of states at every time. The resulting law converges within its parity-compatible half at exact total-variation distance  $1/(2^d(2^d-1)^{t-1})$ ; it does not mix in ordinary total variation because the chain has period two. On the whole carrier the only eigenvalues are  $1, -1, \pm(2^d-1)^{-1}$ , with all multiplicities explicit and no Jordan blocks. Simplex-code, incidence-design, Cayley/Fourier, and crown-graph facts are treated as background. The residual claim remains OWNER\_AMBER; external circulation is HOLD\_EXTERNAL.

## 1. MODEL, BACKGROUND BOUNDARY, AND THEOREM

Fix  $d \geq 2$ , put  $V = \mathbb{F}_2^d$ ,  $E = V \setminus \{0\}$ , and write

$$(1) \quad q = 2^d, \quad N = q - 1, \quad m = |E| = N.$$

We identify a subset of  $E$  with its incidence word in  $\mathbb{F}_2^E$ ; addition of words is symmetric difference. At each epoch choose, independently and uniformly, a nonzero form  $\ell \in V^* \setminus \{0\}$  and update

$$(2) \quad A \mapsto A + h_\ell, \quad h_\ell = \mathbf{1}_{H_\ell}, \quad H_\ell = \{x \in E : \ell(x) = 0\}.$$

Over  $\mathbb{F}_2$ , the nonzero forms index the projective hyperplanes without a scalar ambiguity. The zero form is not sampled. In particular,  $|H_\ell| = 2^{d-1} - 1$  is odd.

Projective systems and simplex codes are standard coding-theoretic objects [3]; binary codes, symmetric-difference designs, and projective hyperplane sections also have a direct established connection [4]. Walks involving generating sets of Abelian groups provide broader Markov-chain context [2]. Brown's hyperplane-chamber walks [1] are a useful terminology control but have a different state space and update from (2). We assign simplex-code and incidence language, symmetric difference, Cayley-graph reduction, finite-abelian character calculus, the named crown graph, and its elementary spectrum no contribution credit. The calculations below are included to make the literal conjunction self-contained.

For  $a \in V^*$  define the evaluation word

$$(3) \quad c_a = (a(x))_{x \in E}, \quad \mathcal{C} = \{c_a : a \in V^*\}, \quad \mathcal{W} = \langle \mathbf{1}, \mathcal{C} \rangle \leq \mathbb{F}_2^E,$$

where  $\mathbf{1}$  is the all-one word. Let  $P$  be the transition operator of (2) and set

$$(4) \quad K = 2^{m-d-1}.$$

**Theorem 1** (complete finite atlas). *The following statements hold for every  $d \geq 2$ .*

- (i) *The closed communicating classes are precisely the  $K$  cosets of  $\mathcal{W}$ . Each has  $2q$  states, all states are recurrent, and the support on each class is the crown graph  $K_{q,q}$  with one perfect matching deleted. The chain is simple random walk on that graph and has period two.*
- (ii) *For  $t \geq 0$  and  $A, B \subseteq E$ , the endpoint lies in the parity-compatible phase exactly when there is a unique  $L \in V^*$  such that*

$$(5) \quad A + B = (t \bmod 2)\mathbf{1} + c_L.$$

*When this condition holds, the number of ordered form histories is*

$$(6) \quad a_t(L) = \begin{cases} q^{-1}(N^t + N(-1)^t), & L = 0, \\ q^{-1}(N^t - (-1)^t), & L \neq 0, \end{cases} \quad P^t(A, B) = \frac{a_t(L)}{N^t}.$$

*A history exists exactly when (5) holds and  $a_t(L) > 0$ : this requires  $L = 0$  at  $t = 0$ ,  $L \neq 0$  at  $t = 1$ , and permits every  $L$  at  $t \geq 2$ . Outside (5),  $P^t(A, B) = 0$ .*

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(iii) Put

$$(7) \quad \Phi_t(A) = A + (t \bmod 2)\mathbf{1} + C$$

and let  $U_{\Phi_t(A)}$  be uniform on these  $q$  states. For  $t \geq 1$ ,

$$(8) \quad \|P^t(A, \cdot) - U_{\Phi_t(A)}\|_{\text{TV}} = \frac{1}{qN^{t-1}}.$$

If  $\pi_{A+W}$  is uniform on the full communicating class, then

$$(9) \quad \|P^t(A, \cdot) - \pi_{A+W}\|_{\text{TV}} = \begin{cases} \frac{1}{2} + \frac{1}{2q}, & t = 1, \\ \frac{1}{2}, & t \geq 2. \end{cases}$$

Thus (8) is parity-phase convergence, not ordinary mixing.

(iv) On the entire  $2^m$ -state carrier, the eigenvalues and algebraic multiplicities are

$$(10) \quad \begin{array}{c|cccc} \lambda & 1 & -1 & 1/N & -1/N \\ \hline \text{mult}(\lambda) & K & K & NK & NK. \end{array}$$

Boolean characters give a complete eigenbasis. Hence there are no other eigenvalues and no non-trivial Jordan blocks.

(v) Within this family, the degree  $N$  of one unlabelled component recovers  $d = \log_2(N + 1)$ . If the total state count is also supplied, then it recovers  $K$  as the number of components. This is not a characterization among arbitrary regular bipartite graphs.

For reference, the theorem's quantitative spine is

| object              | exact value                                           |
|---------------------|-------------------------------------------------------|
| class decomposition | $K = 2^{m-d-1}$ cosets, each of size $2q$             |
| component support   | $K_{q,q}$ minus a matching; degree $N$ ; period 2     |
| endpoint kernel     | (5)–(6)                                               |
| phase convergence   | $1/(qN^{t-1})$ in total variation                     |
| global spectrum     | $1, -1, 1/N, -1/N$ with multiplicities $K, K, NK, NK$ |

## 2. CODE COORDINATES AND CROWN COMPONENTS

**Lemma 2.** The map  $a \mapsto c_a$  is injective and linear,  $\mathbf{1} \notin \mathcal{C}$ , and  $\dim \mathcal{W} = d + 1$ . Moreover, the masks  $h_\ell$ ,  $\ell \neq 0$ , generate  $\mathcal{W}$ .

*Proof.* Linearity is coordinatewise. If  $a \neq 0$ , some nonzero  $x$  satisfies  $a(x) = 1$ , so  $c_a \neq 0$  and the evaluation map is injective. A nonzero form has a kernel of dimension  $d - 1 \geq 1$ ; hence it cannot equal one on every nonzero vector. Thus  $\mathbf{1} \notin \mathcal{C}$ , proving  $\dim \mathcal{W} = d + 1$ .

For every nonzero  $\ell$  and  $x \in E$ ,

$$(11) \quad h_\ell(x) = 1 + \ell(x), \quad h_\ell = \mathbf{1} + c_\ell.$$

Given  $u \neq 0$ , choose  $a \notin \{0, u\}$ , which is possible because  $d \geq 2$ , and put  $b = a + u$ . Then  $a, b$  are nonzero and  $h_a + h_b = c_{a+b} = c_u$ . Finally  $\mathbf{1} = h_u + c_u$ , so the masks generate both  $\mathcal{C}$  and  $\mathbf{1}$ .  $\square$

*Proof of Theorem 1(i).* A walk on an additive group remains in the coset of the subgroup generated by its increments. Lemma 2 identifies that subgroup as  $\mathcal{W}$ , so its cosets are exactly the closed irreducible classes. Their size is  $2^{d+1} = 2q$ , and (4) gives their number. Finite closed irreducible classes contain only recurrent states.

Fix a coset representative  $R$ . The unique code coordinates give a bijection

$$(12) \quad R + \epsilon \mathbf{1} + c_a \longleftrightarrow (\epsilon, a), \quad (\epsilon, a) \in \mathbb{F}_2 \times V^*.$$

By (11), a step indexed by  $\ell \neq 0$  sends  $(\epsilon, a)$  to  $(\epsilon + 1, a + \ell)$ . It therefore joins  $(0, a)$  to every  $(1, b)$  except  $b = a$ , exactly the crown graph  $K_{q,q}$  minus its diagonal matching. Every edge changes subset parity because every mask has odd weight. Thus all returns have even length, while repeating any mask gives a two-step return. The period is exactly two.  $\square$

## 3. ORDERED HISTORIES AND THE TWO TV COMPARISONS

After the ordered choices  $\ell_1, \dots, \ell_t$ , linearity in (11) gives the total increment

$$(13) \quad \sum_{r=1}^t h_{\ell_r} = (t \bmod 2)\mathbf{1} + c_L, \quad L = \sum_{r=1}^t \ell_r.$$

Injectivity of  $a \mapsto c_a$  proves the endpoint condition and uniqueness in Theorem 1(ii).

*Proof of (6).* The additive characters of  $V^*$  give the exact number of ordered nonzero summands with sum  $L$ :

$$(14) \quad a_t(L) = \frac{1}{q} \sum_{y \in V^{**}} (-1)^{y(L)} \left( \sum_{\ell \neq 0} (-1)^{y(\ell)} \right)^t.$$

The inner sum is  $N$  for the trivial character and  $-1$  otherwise. The sum of all nontrivial characters at  $L$  is  $N$  when  $L = 0$  and  $-1$  otherwise. This proves (6), including  $t = 0$ . Division by the  $N^t$  equiprobable histories proves the transition formula. Directly in (6),  $a_0(L)$  is positive only for  $L = 0$ ,  $a_1(L)$  is positive only for  $L \neq 0$ , and both displayed values are positive for every  $L$  when  $t \geq 2$ . These are precisely the asserted support cases.  $\square$

*Proof of Theorem 1(iii).* On the  $q$ -point set (7), the probability deviations from  $1/q$  are

$$(15) \quad p_t(0) - \frac{1}{q} = \frac{(-1)^t}{qN^{t-1}}, \quad p_t(L) - \frac{1}{q} = -\frac{(-1)^t}{qN^t} \quad (L \neq 0).$$

Half the sum of their absolute values is  $1/(qN^{t-1})$ , proving (8).

The connected regular crown walk has the uniform stationary law on all  $2q$  component vertices. At  $t = 1$ , one point in the occupied parity half has mass zero and the other  $N$  points have mass  $1/N$ ; direct subtraction from  $1/(2q)$  gives the first line of (9). For  $t \geq 2$ , (15) and  $N \geq 3$  show that every probability in the occupied half exceeds  $1/(2q)$ . That half contributes total absolute deviation  $1/2$ , and the empty half contributes another  $1/2$ . Dividing by two proves the second line. In particular, the phase limit must not be reported as ordinary mixing.  $\square$

## 4. FULL-CARRIER SPECTRUM AND RECONSTRUCTION

For  $S \subseteq E$ , define the Boolean character

$$(16) \quad \chi_S(A) = (-1)^{|S \cap A|}, \quad \sigma(S) = \sum_{x \in S} x \in V.$$

These  $2^m$  characters form an orthogonal basis on  $\mathbb{F}_2^E$ .

*Proof of Theorem 1(iv).* Convolution by the uniform mask measure makes  $\chi_S$  an eigenvector with eigenvalue

$$(17) \quad \lambda_S = \frac{1}{N} \sum_{\ell \neq 0} (-1)^{|S \cap H_\ell|}.$$

Modulo two,  $|S \cap H_\ell| = |S| + \ell(\sigma(S))$ . Character orthogonality therefore gives

$$(18) \quad \lambda_S = \begin{cases} (-1)^{|S|}, & \sigma(S) = 0, \\ (-1)^{|S|+1}/N, & \sigma(S) \neq 0. \end{cases}$$

It remains to count the four cases without hiding a rank assumption. The linear map

$$(19) \quad \mathbb{F}_2^E \longrightarrow \mathbb{F}_2 \oplus V, \quad S \longmapsto (|S| \bmod 2, \sigma(S))$$

has columns  $(1, x)$  for  $x \in E$ . For any  $u \neq 0$ , choose  $x \notin \{0, u\}$ ; then  $(1, x) + (1, x+u) = (0, u)$ . These vectors span  $\{0\} \oplus V$ , and  $(1, x) + (0, x) = (1, 0)$  supplies the remaining direction. The map is surjective, so every fibre has size  $2^{m-d-1} = K$ . Sorting (18) by parity and by whether  $\sigma(S)$  vanishes proves (10). The full character basis also proves diagonalizability, so no Jordan block is concealed by the repeated multiplicities.  $\square$

*Proof of Theorem 1(v).* The crown support has degree  $N$  because the  $N$  nonzero forms give distinct masks. Hence  $N + 1 = 2^d$  recovers  $d$ . A component has  $2(N + 1) = 2q$  states; dividing the supplied total state count by this number recovers  $K$ . The argument uses the promised family and makes no inverse claim for an arbitrary graph with the same degree and order.  $\square$

## 5. EXACT CONTROLS, EXCLUSIONS, AND STATUS

The paper-local author-side verifier uses only integer and rational arithmetic and imports no scouting or prior-paper code. It checks 1,095,999 exact assertions: literal carriers, cosets, neighborhoods, and Boolean characters through  $d = 4$ ; ordered histories through five steps in those boxes; and the algebraic formulas through  $d = 8$  and sixteen steps. This is falsification pressure, not a proof or an owner-search instrument.

At  $d = 1$ , the sole projective hyperplane is empty, so the chain is the identity rather than a crown walk. Sampling the zero form for  $d \geq 2$  would add an empty mask; toggling hyperplane complements would replace the generators  $\mathbf{1} + c_\ell$  by  $c_\ell$ . Neither variant is covered. No claim is made for nonbinary spaces.

The remaining claim is deliberately owner-thin: the literal process plus the disjoint-crown conjugacy, exact labelled history kernel, parity-phase TV, and full-carrier multiplicity lift. Generic Fourier and the finite-group proof shell already appear in the internal P145 line and receive zero separation credit. A bounded literature non-hit is not a novelty result. The author-side gate is OWNER\_\_AMBER, and the external lifecycle is HOLD\_\_EXTERNAL.

## REFERENCES

1. Kenneth S. Brown, *Semigroups, rings, and markov chains*, Journal of Theoretical Probability **13** (2000), no. 3, 871–938.
2. P. Diaconis and L. Saloff-Coste, *Walks on generating sets of abelian groups*, Probability Theory and Related Fields **105** (1996), no. 3, 393–421.
3. Mariusz Kwiakowski, Mark Pankov, and Antonio Pasini, *The graphs of projective codes*, Finite Fields and Their Applications **54** (2018), 15–29.
4. Vladimir D. Tonchev, *Quasi-symmetric designs, codes, quadrics, and hyperplane sections*, Geometriae Dedicata **48** (1993), no. 3, 295–308.